

On the Collisional Ordering of a Shear-Flow Stabilized Cubic, Pureflow Bennett-Shumlak-Hartman Vortex

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This brief note shows that the ion fluid momentum reduces to the Ideal Ohm's Law when the standard ordering $\omega_{pe} \gg \bar{v}_{ei}$ holds for the non-relativistic Braginskii Coulomb collision frequency.

The ion momentum equation reads,

$$\begin{aligned} m_i n_i \left(\frac{\partial}{\partial t} + \vec{u}_i \cdot \nabla \right) \vec{u}_i \\ = e n_i (\vec{E} + \vec{u}_i \times \vec{B}) - \nabla p_i - m_e n_e \bar{v}_{ei} (\vec{u}_i - \vec{u}_e) \end{aligned} \quad (1)$$

where the equilibrium-averaged Braginskii electron-ion collision frequency,

$$\bar{v}_{ei} = (2.9 * 10^{-6}) \frac{n_e \ln(\Lambda)}{T_e^{3/2}} \quad (2)$$

Unlike in the electron case no ad hoc relativistic modifications are required to obtain the Ideal Ohm's Law, only the standard non-relativistic plasma ordering,

$$\omega_{pe} \gg \bar{v}_{ei} \quad (3)$$

and the Z-pinch ansatz,

$$\vec{u}_i = u_i(r) \hat{z} \quad (4)$$

The electron plasma frequency is,

$$\omega_{pe}^2 = \frac{n_e e^2}{m_e \epsilon_0} \quad (5)$$

so that with quasineutrality,

$$n_e = n_i \quad (6)$$

we have for the ion momentum equation,

$$\begin{aligned} m_i n_e \left(\frac{\partial}{\partial t} + \vec{u}_i \cdot \nabla \right) \vec{u}_i \\ = e n_e (\vec{E} + \vec{u}_i \times \vec{B}) - \nabla p_i - m_e n_e \bar{v}_{ei} (\vec{u}_i - \vec{u}_e) \end{aligned} \quad (7)$$

which becomes,

$$\begin{aligned} \therefore m_i \frac{m_e \epsilon_0}{e^2} \omega_{pe}^2 \left(\frac{\partial}{\partial t} + \vec{u}_i \cdot \nabla \right) \vec{u}_i \\ = \frac{m_e \epsilon_0}{e} \omega_{pe}^2 (\vec{E} + \vec{u}_i \times \vec{B}) - \nabla p_i - \frac{m_e^2 \epsilon_0}{e^2} \omega_{pe}^2 \bar{v}_{ei} (\vec{u}_i - \vec{u}_e) \end{aligned} \quad (8)$$

(9)

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Dividing both sides by $\frac{1}{\omega_{pe}^3}$, we obtain,

$$\therefore m_i \frac{m_e \epsilon_0}{e^2} \frac{1}{\omega_{pe}} \left(\frac{d}{dt} + \vec{u}_i \cdot \nabla \right) \vec{u}_i \quad (10)$$

$$= \frac{m_e \epsilon_0}{e} \frac{1}{\omega_{pe}} (\vec{E} + \vec{u}_i \times \vec{B}) - \frac{1}{\omega_{pe}^3} \nabla p_i - \frac{m_e^2 \epsilon_0}{e^2} \frac{\bar{v}_{ei}}{\omega_{pe}} (\vec{u}_i - \vec{u}_e) \quad (11)$$

the collisional term then disappears because of the aforementioned non-relativistic ordering. Similarly, the inertial forces on the LHS will as well because of the Z-pinch ansatz. This leaves,

$$\frac{m_e \epsilon_0}{e} \frac{1}{\omega_{pe}} (\vec{E} + \vec{u}_i \times \vec{B}) - \frac{1}{\omega_{pe}^3} \nabla p_i = 0 \quad (12)$$

$$\rightarrow \frac{m_e \epsilon_0}{e} \frac{1}{\omega_{pe}} (\vec{E} + \vec{u}_i \times \vec{B}) = \frac{1}{\omega_{pe}^3} \nabla p_i \quad (13)$$

Let us multiply through by $\omega_{pe}^2 \bar{v}_{ei}$, obtaining,

$$\frac{m_e \epsilon_0}{e} \omega_{pe} \bar{v}_{ei} (\vec{E} + \vec{u}_i \times \vec{B}) = \frac{\bar{v}_{ei}}{\omega_{pe}} \nabla p_i \quad (14)$$

so that for sufficiently small pressure gradients we can eliminate the term on the RHS asymptotically, yielding,

$$\frac{m_e \epsilon_0}{e} \omega_{pe} \bar{v}_{ei} (\vec{E} + \vec{u}_i \times \vec{B}) = 0 \quad (15)$$

which requires,

$$\vec{E} = -\vec{u}_i \times \vec{B} \quad (16)$$

linking the MHD flow with the two-fluid drift velocity then requires stationary electrons because for non-relativistic plasmas the MHD flow is naturally very close to the ion flow,

$$\vec{u} \simeq \vec{u}_i \quad (17)$$

and with stationary electrons the same is true for the two-fluid drift,

$$\vec{U} = \vec{u}_i - \vec{u}_e \simeq \vec{u}_i \quad (18)$$

the physical picture behind this is that when the electrons are oscillating extremely fast about a central point, then from the perspective of the ions their motion is so fast that in an average sense do not appear to move from this central point.

